



Chemical kinetics study of 1,3-butadiene + HO₂; implications for combustion modeling and simulation

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ABSTRACT

Detailed reaction kinetics for the system of 1,3-butadiene + hydroperoxyl radical (HO₂) and its application in combustion modeling have been investigated in this work. Zero-point energy (ZPE) corrected potential energy surfaces (PESs) for the title reactions have been obtained using high-level ab initio electronic structure methods of ROCCSD(T)/CBS//BHandHLYP/6–311++G(d,p), and the energies relative to the reactants for all stationary points were used in the kinetic calculations. Pressure- and temperature-dependent rate constants for each individual reaction pathway have been calculated by solving the Rice-Ramsperger-Kassel-Marcus/Master Equation (RRKM/ME) implemented in the Master Equation System Solver (MESS) program. The bimolecular pathways forming vinyl oxirane, (CYOC)C=C, and hydroxyl radical (OH) from the reactants are found to be dominant in the temperature range of 800–2000 K below 1 atm, and keep competitive when temperature is higher than 1000 K at 100 atm. At elevated pressures, the stabilizations of the two entrance channels forming radicals of 1-hydroperoxy-3-buten-2-yl (C₄H₆1–3OOH4) and 3-butenyl-2-peroxy (C₄H₇1–OO3) are more important than the formation of bimolecular products (CYOC)C=C + OH at temperatures below 1000 K. Temperature-dependent thermochemical properties for important species on the PESs of 1,3-butadiene + HO₂ have also been calculated. The kinetics and thermochemistry data for the title reaction system calculated in this work have been incorporated into AramcoMech 3.0 mechanism and their influence on the performance of AramcoMech 3.0 in simulating the combustion properties of 1,3-butadiene, including ignition delay times and species profiles against temperature or time, has been investigated. This is a first-time study on the reaction kinetics between HO₂ radical and species including two double bonds.

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1. Introduction

1,3-butadiene (CH₂=CHCH=CH₂, C₄H₆) is an important unsaturated intermediate in the pyrolysis and oxidation of higher-order hydrocarbons. It can be produced in significant amount by other unsaturated species such as 1-butene [1], 2-butene [2] and pentene isomers [3], etc. Also, it was found to be the major product of reactions involving larger diene, for example, H-atom addition to 1,3-pentadiene can lead to significant yields of 1,3-butadiene [4]. Moreover, 1,3-butadiene has long been recognized as an important precursor of polycyclic aromatic hydrocarbons (PAHs) [5] given that the resonantly stabilized C₄H₅ radicals produced from 1,3-butadiene oxidation can directly form “the first aromatic ring”, benzene, via “C₄ + C₂” addition reactions. And a recent modeling study of sooting ethylene premixed flames [6] suggested the need for further investigating 1,3-butadiene oxidation to account

for its consumption and the formation of other benzene precursors related to it (e.g. allene) more accurately. Hence, improved knowledges of 1,3-butadiene oxidation kinetics are important not only for understanding the oxidation mechanisms of unsaturated hydrocarbons but also for the PAH chemistry.

A number of experimental and modeling studies have been conducted on 1,3-butadiene chemistry. Laskin et al. [7] measured concentration profiles for some C₁–C₄ species in 1,3-butadiene pyrolysis and oxidation over temperatures in the range of 1035–1185 K at 1 atm, using turbulent flow reactor. Libby et al. [8] conducted hydroxyl (OH) radical time history measurement while measuring ignition delay time for 1,3-butadiene oxidation in shock tube at 1390–1810 K and 1.4–3.9 atm with equivalence ratio at 0.25, 0.5 and 1.0. More recently, a detailed chemical kinetic mechanism (AramcoMech 3.0) capable of describing 1,3-butadiene combustion was proposed by Zhou et al. [9] and was validated against ignition delay time (IDT), laminar flame speed and jet-stirred reactor and flow reactor speciation measurements fueling by 1,3-butadiene under broad range of temperatures, pressures and equivalence ratios, including those in Refs. [7,8]. Hydroperoxyl (HO₂)

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radical addition to 1,3-butadiene and their subsequent reactions were found to be pivotal in predicting the IDTs at temperature in the range from 600 to 900 K. However, no experimental or theoretical kinetics data for the reactions of 1,3-butadiene + $\text{H}\dot{\text{O}}_2$ are available in the literature. The rate constants implemented in the AramcoMech 3.0 mechanism for the related reaction pathways were defined based on the high pressure limit rate constants for the analogous pathways in 1-butene + $\text{H}\dot{\text{O}}_2$ [9], adjusting the pre-exponential factors by a factor of two to take the reaction path degeneracy into consideration.

Rate constants of the association reactions of $\text{H}\dot{\text{O}}_2$ and the alkenes with one double bond have been studied by different groups [10,11]. Hydroperoxyl radicals add to the double bond of alkene through concerted addition and direct addition, forming alkyl peroxy radical ($\text{ROO}\cdot$) and hydroperoxyl alkyl radical ($\cdot\text{QOOH}$) respectively, are two distinct entrance reaction pathways of alkene + $\text{H}\dot{\text{O}}_2$. The $\text{ROO}\cdot$ radical can decompose to alkyl radical and O_2 , while the $\cdot\text{QOOH}$ radical can undergo cyclization reaction to produce cyclic ether and $\dot{\text{O}}\text{H}$ radical. Zádor et al. [10] systematically investigated the kinetics of $\text{H}\dot{\text{O}}_2$ adding to seven unsaturated species including ethylene, propene, three butene isomers, cyclohexene and vinyl alcohol. In their study, the formation of cyclic ether + $\dot{\text{O}}\text{H}$ is prevalent due to the relatively shallow well of $\cdot\text{QOOH}$, and the formations of $\text{ROO}\cdot$ and $\cdot\text{QOOH}$ were found to be important at elevated pressure. Villano et al. [11] investigated the rate rules of $\text{H}\dot{\text{O}}_2$ adding to a series of alkenes from C_2 – C_7 with different degrees of branching, and unraveled the competition between the formation of cyclic ether + $\dot{\text{O}}\text{H}$ and the stabilization of $\cdot\text{QOOH}$ at higher pressures. In addition, reaction systems of allyl + O_2 explored by Lee and Bozzelli [12] and of 3-butenyl + O_2 studied by Miyoshi [13] also provide useful information for the investigation of diene + $\text{H}\dot{\text{O}}_2$ sub mechanism, since the potential energy surfaces of these two reaction systems can both be entered from diene (allene and 1,3-butadiene, respectively) + $\text{H}\dot{\text{O}}_2$ as well.

In this study, the comprehensive zero-point energy corrected potential energy surfaces of 1,3-butadiene + $\text{H}\dot{\text{O}}_2$ have been explored for the first time, using high-level electronic structure methods. The pressure- and temperature-dependent rate coefficients have been obtained from the solution of RRKM/ME. Temperature-dependent thermochemistry data for the important species have been obtained from ab initio calculations. The calculated kinetics and thermochemistry data have been incorporated into the AramcoMech 3.0 mechanism to simulate their influence on ignition delay times. Rate of production analyses have been performed to analyze the difference arises from 1,3-butadiene + $\text{H}\dot{\text{O}}_2$ chemistry.

2. Computational methods

The optimized geometries, vibrational frequencies and zero-point energy (ZPE) corrections for all species and transition states (TS) involved were obtained at the BHandHLYP/6–311++G(d,p) [14] level of theory. The intrinsic reaction coordinate (IRC) calculations were carried out at the same level of theory to confirm the TSs connecting with the desired reactants and products. One-dimensional hindered rotor treatments were employed for low-frequency torsional modes that are related to the C–C, C–O and O–O single bonds, and the relaxed PESs were scanned in 5-degree increments using BHandHLYP/6–31G(d) methodology to obtain the energies as a function of dihedral angle. The corresponding structure of global minimum on the rotational potentials of each individual stationary point (if it has internal rotations) was used as its input geometry for single-point energy (SPE) calculations.

The restricted open-shell coupled cluster method ROCCSD(T) [15] along with the cc-pVTZ and cc-pVQZ [16,17] basis sets were used in the SPE calculation for each stationary point, and the SPEs

were extrapolated to the complete basis set (CBS) limit [18,19]:

$$E_{\text{ROCCSD(T)/CBS}} = E_{\text{ROCCSD(T)/cc-pVQZ}} + (E_{\text{ROCCSD(T)/cc-pVQZ}} - E_{\text{ROCCSD(T)/cc-pVTZ}})4^4/(5^4 - 4^4) \quad (1)$$

The T1 diagnostics [20] at the ROCCSD(T)/cc-pVQZ level of theory for all species are below 0.025, except for two radicals (both are 0.032), while those for all TSs are no larger than 0.031, except for two (0.036 and 0.044). According to the T1 diagnostic values for species and most of TSs, single-reference methods can be used in SPE calculations, as suggested by Goldsmith et al. [21]. The influence of this treatment, for the two TSs with high T1 diagnostic values, on the kinetics of title reaction system is discussed in Section 3. Moreover, five other methods including G3 [22], G4 [23], CBS-APNO [24], CBS-QB3 [25,26], and a combination of CCSD(T)/aug-cc-pVTZ, MP2/aug-cc-pVTZ and MP2/aug-cc-pVQZ [27] methodologies:

$$E_{\text{CCSD(T)/MP2/CBS}} = E_{\text{CCSD(T)/aug-cc-pVTZ}} + (E_{\text{MP2/aug-cc-pVQZ}} - E_{\text{MP2/aug-cc-pVTZ}}) \quad (2)$$

were employed to compute the ZPE corrected electronic energies for the important entrance wells and the corresponding TSs for comparison. Gaussian 09 [28] and Molpro 2015 [29] programs were used to perform the high-level ab initio calculations.

The RRKM/ME was solved by Master Equation System Solver (MESS) code [30,31] to obtain the pressure- and temperature-dependent rate constants at temperatures in the range of 298.15–2000 K at pressures of 0.01, 0.1, 1, 10 and 100 atm respectively, for all reaction pathways displayed in Fig. 1(a) and (b) in N_2 bath. The collisional energy transfer was described by an exponential down model and the average energy in deactivating collision was estimated by $\langle \Delta E_{\text{down}} \rangle = 200(T/300)^{0.75} \text{ cm}^{-1}$, which yields reasonable results for C_4 hydrocarbons as Li et al. discussed in [32]. Phase space theory [33] was used to calculate rate constants for the barrierless bond fission channels of the $\text{ROO}\cdot$ intermediates that forming butenyl + O_2 . The potential pre-factor (C_n) and potential power exponent (n) in the potential energy expression $V = -C_n/R^n$ as a function of distance (R) in the implementation of phase space theory to estimate number of states, for $\text{C}_4\text{H}_7\text{-1-OO3} \rightarrow \text{C}_4\text{H}_7\text{-1-3} + \text{O}_2$, were adjusted to match the high-pressure limiting (HPL) rate constants of the association of $\text{C}_4\text{H}_7\text{-1-3}$ and O_2 with the presumed ones, which were estimated to be equal to the HPL rate constants for benzyl + $\text{O}_2 \rightarrow \text{C}_7\text{H}_7\text{O}_2$ calculated by Pelucchi et al. [34], given that the two processes have similar allylic C–O bonds and bond dissociation energies of $\sim 20 \text{ kcal/mol}$. And the two parameters for $\text{C}_4\text{H}_7\text{-1-OO4} \rightarrow \text{C}_4\text{H}_7\text{-1-4} + \text{O}_2$ were adjusted to match the HPL rate constants for $\text{C}_4\text{H}_7\text{-1-4} + \text{O}_2 \rightarrow \text{C}_4\text{H}_7\text{-1-OO4}$ and the HPL rate constants for 1-butyl + $\text{O}_2 \rightarrow \text{C}_4\text{H}_9\text{OO}$ calculated by Miyoshi [35]. Output files for the calculations in MESS package are provided in the Supplementary material.

Enthalpies of formation at 0 K for the important species in reactions of 1,3-butadiene + $\text{H}\dot{\text{O}}_2$ were determined based on atomization approach, by the average of atomization formation enthalpies calculated from composite methods G3 [22], G4 [23], CBS-QB3 [24] and CBS-APNO [25,26]. Simmie and Somers [36] proved the statistical advantages of such combination and showed that this cost-effective alternative to high-level methods reproduced the benchmark formation enthalpies in AtcT database to within $\sim 1 \text{ kcal/mol}$, rivaling “chemical accuracy” ($\pm 4.184 \text{ kJ/mol}$). Then the molecular partition functions computed by the MESS code [30,31] based on the optimized geometries and vibrational frequencies from the BHandHLYP/6–311++G(d,p) method with 1-D hindered rotors included, along with the 0 K formation enthalpies, were used to calculate temperature-dependent enthalpies of formation, entropies and heat capacities with the ThermP program [37,38], and they were fitted to NASA polynomials by the PAC99 code [37–39].

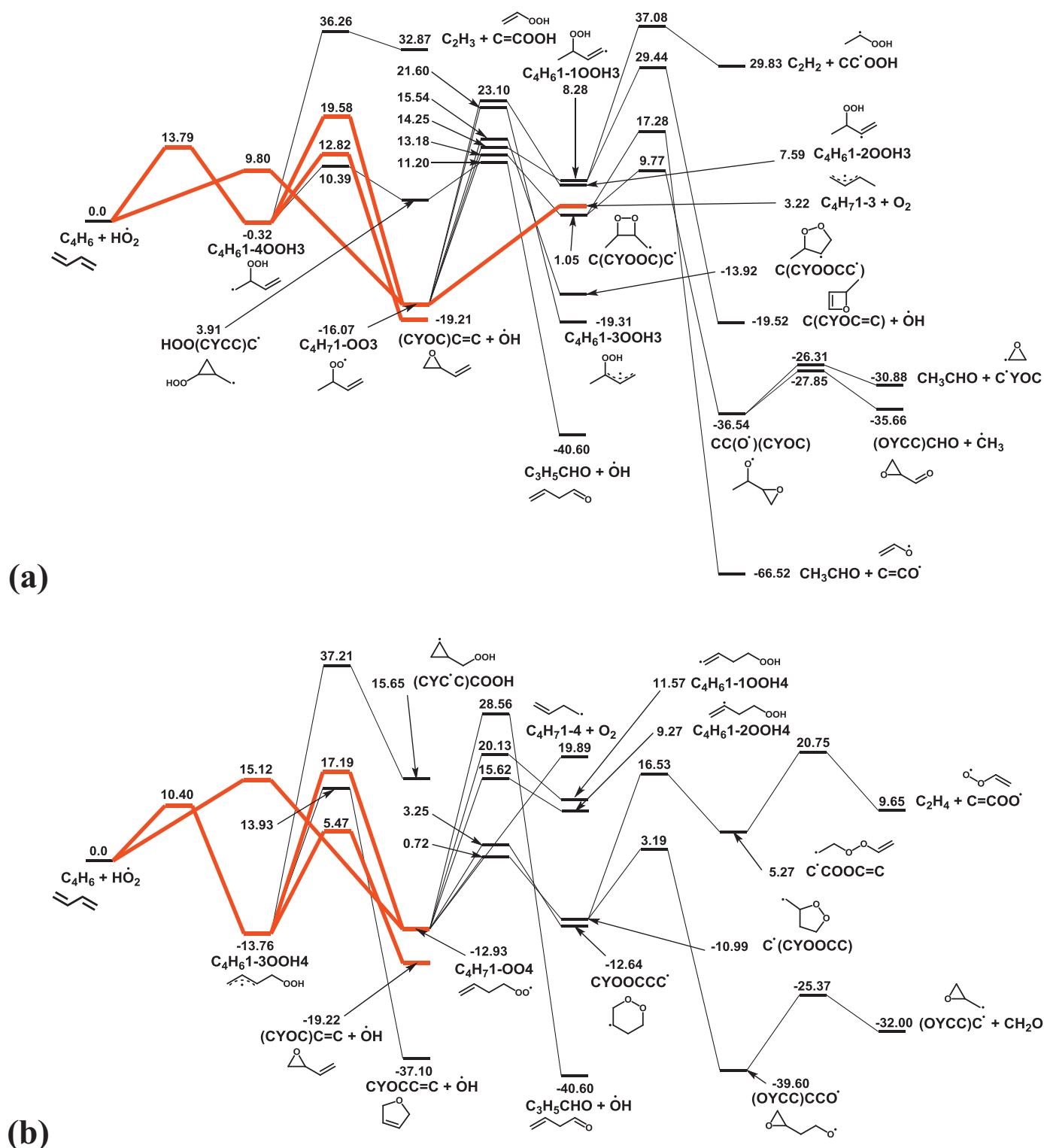


Fig. 1. ZPE corrected potential energy surfaces obtained at ROCCSD(T)/CBS//BHandHLYP/6–311++G(d,p) level of theory for the reaction system of 1,3-butadiene + HO_2 . (a) Central addition reaction pathways, (b) terminal addition reaction pathways, with energies in units of kcal/mol relative to the reactants.

3. Results and discussion

3.1. Reaction pathways and potential energy surfaces

The ZPE corrected PESs at ROCCSD(T)/CBS//BHandHLYP/6–311++G(d,p) level of theory for the detailed reaction pathways of 1,3-butadiene + HO_2 are illustrated in Fig. 1. Hydroperoxyl radical

can add to the central or terminal carbon atom in 1,3-butadiene directly, forming 2-hydroperoxy-3-buten-1-yl ($C_4H_6-1-4OOH3$) shown in Fig. 1(a), and resonantly stabilized 1-hydroperoxy-3-buten-2-yl ($C_4H_6-1-3OOH4$) radicals shown in Fig. 1(b). Alternatively, the concerted addition reaction can also happen, in which the H atom and OO group in HO_2 radical add to the two ends of the double bond in C_4H_6 separately with O–H bond cleavage in HO_2 radical, forming

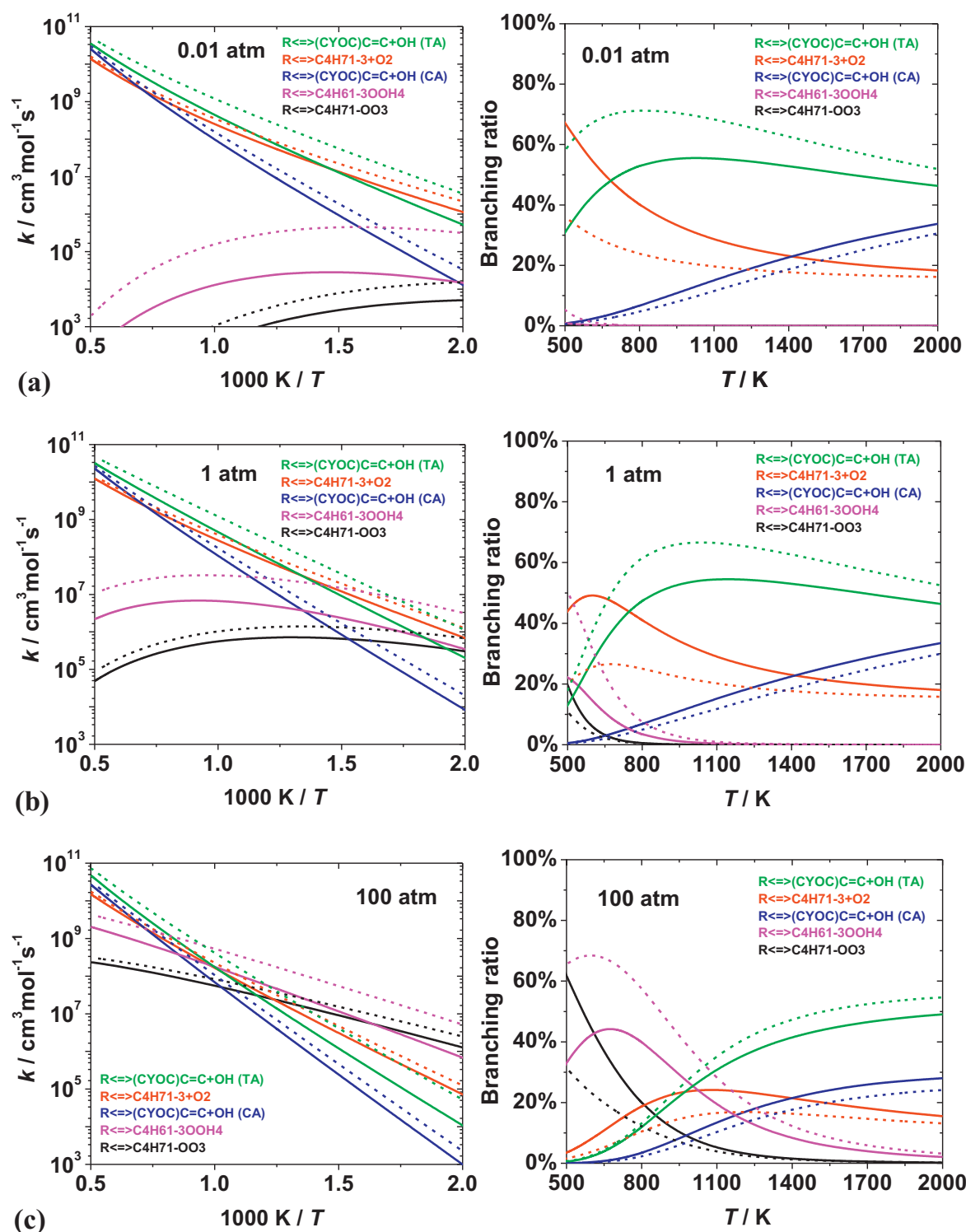


Fig. 2. Kinetic calculation results of HO_2 addition to 1,3-butadiene. (a) Pressure- and temperature-dependent rate constants for the important reaction pathways at 0.01, 1 and 100 atm. (b) Branching ratios of the important reaction pathways at 0.01, 1 and 100 atm. (Solid lines: barrier heights obtained from method of ROCCSD(T)/CBS were employed; dashed lines: barrier heights obtained from method of CCSD(T)/MP2/CBS were employed; R refers to the reactants of $\text{C}_4\text{H}_6 + \text{HO}_2$; TA and CA refer to terminal addition channels and central addition channels, respectively.).

3-butenyl-2-peroxy ($\text{C}_4\text{H}_7\text{1-OO3}$) radical when the OO group adds to the central carbon atom, and 3-butenyl-1-peroxy ($\text{C}_4\text{H}_7\text{1-OO4}$) radical when the OO group adds to the terminal carbon atom, as illustrated in Fig. 1(a) and (b), respectively. The formation of $\text{C}_4\text{H}_7\text{1-OO3}$ and $\text{C}_4\text{H}_6\text{1-3OOH4}$ are the most important entrance channels for 1,3-butadiene + HO_2 , since barrier heights for these two path-

ways are ~ 4 kcal/mol lower than the barrier heights for the other two entrance channels.

The butenyl peroxy radical $\text{C}_4\text{H}_7\text{1-OO3}$ shown in Fig. 1(a) can isomerize to four hydroperoxyl butenyl radicals $\text{C}_4\text{H}_6\text{1-1OOH3}$, $\text{C}_4\text{H}_6\text{1-2OOH3}$, $\text{C}_4\text{H}_6\text{1-3OOH3}$ and $\text{C}_4\text{H}_6\text{1-4OOH3}$ respectively, when H atom from different site of the butenyl group abstracted

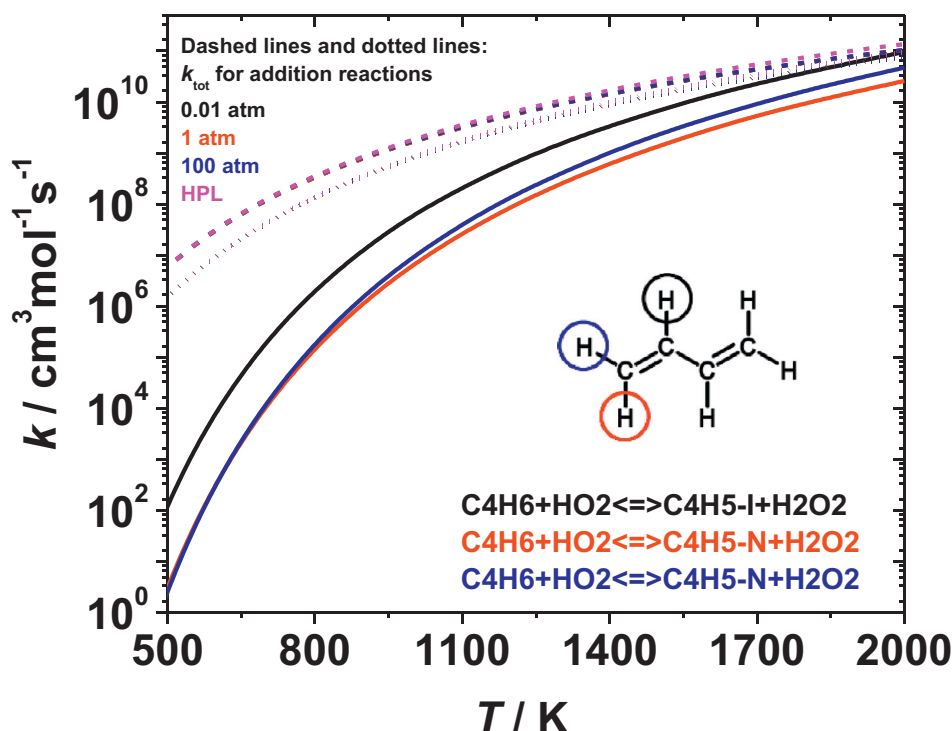


Fig. 3. HPL rate constants for H-atom abstraction reactions from 1,3-butadiene by HO_2 and total rate constants for HO_2 addition to 1,3-butadiene based on CCSD(T)//MP2/CBS barriers (dashed lines) and ROCCSD(T)/CBS barriers (dotted lines).

Table 1

Comparison of the calculated relative energies. R refers to the reactants and W refers to the entrance well. (unit in kcal/mol).

Stationary points	ROCCSD(T)/CBS	CCSD(T)//MP2/CBS	G3	G4	CBS-APNO	CBS-QB3	Miyoshi [13]
TS1: R $\leftrightarrow$ W1	9.8	9.1	10.8	9.3	5.7	9.6	/
TS2: R $\leftrightarrow$ W2	13.8	12.7	13.3	13.0	12.2	10.8	/
TS3: W1 $\leftrightarrow$ W2	19.6	18.8	20.0	17.2	17.6	16.5	/
W1: $\text{C}_4\text{H}_7\text{-OO3}$	-16.1	-16.6	-17.3	-18.4	-17.7	-17.6	/
W2: $\text{C}_4\text{H}_6\text{-4OOH3}$	-0.3	-0.1	/	-1.5	0.0	-1.0	/
TS4: R $\leftrightarrow$ W3	15.1	14.6	/	13.0	12.9	13.4	13.1
TS5: R $\leftrightarrow$ W4	10.4	8.4	8.1	8.7	6.8	5.9	5.7
TS6: W3 $\leftrightarrow$ W4	17.2	15.7	16.7	15.3	13.2	12.8	12.0
W3: $\text{C}_4\text{H}_7\text{-OO4}$	-13.0	-13.5	-14.1	-14.6	-14.5	-14.4	-14.6
W4: $\text{C}_4\text{H}_6\text{-3OOH4}$	-13.8	-13.9	-14.5	-14.9	-14.6	-15.8	-15.8

by the peroxy radical. Energy barrier for the formation of $\text{C}_4\text{H}_6\text{-1-OOH3}$ through a six-membered-ring TS from $\text{C}_4\text{H}_7\text{-OO3}$ is the lowest (30.3 kcal/mol), and barrier heights for the other three pathways are at least 5.3 kcal/mol higher. The peroxy radical in $\text{C}_4\text{H}_7\text{-OO3}$ can also add to the double bond of butenyl group to form cyclic peroxide alkyl radicals $\text{C}(\text{CYOOC})\text{C}\cdot$ and $\text{C}(\text{CYOOC})\cdot$, respectively. The decomposition reaction pathway forming $\text{C}_4\text{H}_7\text{-1-OO3} + \text{O}_2$ is the most important unimolecular pathway of $\text{C}_4\text{H}_7\text{-1-OO3}$, while energy barriers for the other unimolecular pathways are at least 10 kcal/mol higher. This is mainly because the C–O bond to be broken is an allylic-type at the β site relative to O• radical on $\text{C}_4\text{H}_7\text{-1-OO3}$, which leads to lower bond dissociation energy. On the other hand, due to the low-lying well of $\text{C}_4\text{H}_7\text{-1-OO3}$ (–16.1 kcal/mol relative to the reactants), the stabilization of $\text{C}_4\text{H}_7\text{-1-OO3}$ competes with the O_2 elimination pathway at low temperature and high pressure. The formations of $\text{C}_4\text{H}_6\text{-1-OOH3}$ and $\text{C}(\text{CYOOC})\text{C}\cdot$ from $\text{C}_4\text{H}_7\text{-1-OO3}$ are found to be more favored than other isomerization reaction pathways of $\text{C}_4\text{H}_7\text{-1-OO3}$ in kinetic calculations, thus exit channels for these two intermediates leading to bimolecular products have been investigated.

The other butenyl peroxy radical $\text{C}_4\text{H}_7\text{-1-OO4}$ shown in Fig. 1(b), likewise, can isomerize to four hydroperoxyl butenyl radicals of

$\text{C}_4\text{H}_6\text{-1-1OOH4}$, $\text{C}_4\text{H}_6\text{-1-2OOH4}$, $\text{C}_4\text{H}_6\text{-1-3OOH4}$ and $\text{C}_4\text{H}_6\text{-1-4OOH4}$ (decomposes to $\text{C}_3\text{H}_5\text{CHO} + \text{OH}$ instantaneously), and also can decompose to $\text{C}_4\text{H}_7\text{-1-4} + \text{O}_2$ or undergo cyclization reactions to form cyclic peroxide alkyl radicals, $\text{C}(\text{CYOOC})$ and $\text{CYOOC}\cdot$. It can be seen that energy barriers for the H-atom shift and decomposition pathways of $\text{C}_4\text{H}_7\text{-1-OO4}$ lie significantly above the ones for cyclization reaction pathways (by at least 12 kcal/mol), and because the formation of $\text{C}(\text{CYOOC})$ is found to be the most favored among the isomerization reaction channels of $\text{C}_4\text{H}_7\text{-1-OO4}$ in kinetic analyses, subsequent elimination pathways for $\text{C}(\text{CYOOC})$ have been taken into consideration.

The hydroperoxyl butenyl radical $\text{C}_4\text{H}_6\text{-1-3OOH4}$ illustrated in Fig. 1(b), unlike the terminal-adding $\cdot\text{QOOH}$ product of 1-butene + HO_2 which stays at a relatively shallow well (~5 kcal/mol below the reactants) [10,11], lies 13.8 kcal/mol below the reactants due to resonance stabilization. However, the other hydroperoxyl butenyl radical, $\text{C}_4\text{H}_6\text{-1-4OOH3}$ shown in Fig. 1(a), lies at almost the same level as the reactants. The $\text{C}_4\text{H}_6\text{-1-3OOH4}$ radical can undergo ring closure reactions to produce $(\text{CYOC})\text{C}=\text{C} + \text{OH}$, $\text{CYOCC}=\text{C} + \text{OH}$ or $(\text{CYC}\cdot)\text{COOH}$ radical, among which the formation of $(\text{CYOC})\text{C}=\text{C} + \text{OH}$ is the most important since energy barriers for the other two pathways are 8.5 kcal/mol and 31.7 kcal/mol

Table 2Comparison of calculated $\Delta H_{f,298}^\ominus$ and $S_{298}^\ominus$ with literature data for important species on the PES of 1,3-butadiene + HO_2 .

Species	$\Delta H_{f,298}^\ominus$ (kcal mol ⁻¹)		$S_{298}^\ominus$ (cal mol ⁻¹ K ⁻¹)	
	this study	literature	this study	literature
C ₄ H ₆	27.6	26.7 ^a , 26.5 ± 0.1 ^b , 26.5 ± 1.9 ^c , 26.5 ± 0.9 ^d	66.0	66.6 ^a , 70.1 ^c , 65.8 ± 1.3 ^d
C ₄ H ₇ 1-OO3	12.4	14.0 ^a	89.5	86.6 ^a
C ₄ H ₇ 1-OO4	15.9	14.5 ^a , 14.5 ^c	90.7	91.1 ^a , 89.0 ^c
C ₄ H ₆ 1-4OOH3	30.2	29.8 ^a	90.9	90.6 ^a
C ₄ H ₆ 1-3OOH4	15.8	17.1 ^a	89.3	89.8 ^a
(CYOC)C = C	3.1	3.9 ^a	74.1	74.7 ^a
C ₄ H ₇ 1-3	32.7	31.7 ^a , 32.5 ± 1.9 ^c , 32.9 ± 0.9 ^d	71.3	72.0 ^a , 73.1 ^c , 72.0 ± 1.3 ^d
C ₄ H ₇ 1-4	50.7	48.9 ^a , 48.9 ± 1.9 ^c , 49.8 ± 0.9 ^d	77.9	77.2 ^a , 75.8 ^c , 75.5 ± 1.4 ^d
C ₄ H ₅ -I	76.0	76.3 ^a , 75.3 ± 1.9 ^c , 76.2 ± 0.9 ^d	69.1	69.7 ^a , 69.3 ^c , 69.6 ± 1.2 ^d
C ₄ H ₅ -N	87.4	86.5 ^a , 86.8 ± 1.9 ^c , 86.4 ± 0.9 ^d	68.5	68.4 ^a , 72.5 ^c , 68.3 ± 1.3 ^d

^a AramcoMech 3.0 [9]. (Specifically, data for C₄H₆, C₄H₇1-3 and C₄H₇1-3 denoted with ^a were calculated by Li et al. [32] using CBS-APNO/G3/G4//M06-2X/6-311+G(d,p) method).

^b Active Thermochemical Tables (Branco Ruscic Argonne National Lab) [40].

^c Third Millennium Ideal Gas and Condensed Phase Thermochemical Database for Combustion (Burcat) [42]. (Specifically, data for C₄H₇1-3 and C₄H₇1-4 denoted with ^c were calculated by Miller et al. [43] using G3//B3LYP method, while data for other C₄ species denoted by ^c were calculated by Burcat using G3B3 method).

^d Calculated by Goldsmith et al. [41] using RQCISD(T)/cc-pVT,QZ//B3LYP/6-311+G(d,p) method with bond additivity correction.

Table 3Comparison of calculated $C_p(T)$ with literature data for important species on the PES of 1,3-butadiene + HO_2 .

Species	$C_p(T)$ (cal mol ⁻¹ K ⁻¹)													
	This study							Literature						
	300	400	500	600	800	1000	1500	300	400	500	600	800	1000	1500
C ₄ H ₆	18.0	23.1	27.7	31.6	36.6	40.3	46.0	18.9 ^a 17.8 ^c 18.5 ^d	24.3 ^a 22.2 ^c 24.0 ^d	28.7 ^a 26.2 ^c 28.7	32.1 ^a 29.8 ^c 32.4 ^d	37.2 ^a 35.1 ^c 37.6 ^d	40.7 ^a 39.1 ^c 41.1 ^d	46.3 ^a 45.3 ^c 46.6 ^d
C ₄ H ₇ 1-OO3	26.0	31.3	36.2	40.6	47.1	52.0	59.6	27.6 ^a	34.3 ^a	39.8 ^a	44.3 ^a	50.9 ^a	55.4 ^a	61.7 ^a
C ₄ H ₇ 1-OO4	25.7	30.9	36.0	40.4	47.1	52.0	59.6	26.6 ^a	32.3 ^a	37.2 ^a	41.4 ^a	48.1 ^a	53.1 ^a	60.5 ^a
C ₄ H ₆ 1-4OOH3	30.0	35.2	39.7	43.4	49.0	53.2	60.0	25.7 ^c	31.5 ^c	36.8 ^c	41.4 ^c	48.1 ^c	53.1 ^c	60.5 ^c
C ₄ H ₆ 1-3OOH4	28.8	33.6	38.3	42.4	48.5	53.0	60.1	28.2 ^a	34.8 ^a	40.2 ^a	44.6 ^a	51.1 ^a	55.4 ^a	61.3 ^a
(CYOC)C=C	19.9	25.2	30.1	34.3	40.1	44.4	51.0	27.0 ^a	33.1 ^a	38.2 ^a	42.6 ^a	49.4 ^a	54.3 ^a	61.0 ^a
C ₄ H ₇ 1-3	19.6	24.5	29.1	33.1	39.1	43.7	51.0	19.6 ^a 19.9 ^a 19.4 ^c 19.9 ^d	25.6 ^a 25.0 ^a 24.2 ^c 24.8 ^d	30.6 ^a 29.4 ^a 28.7 ^c 29.3 ^d	34.6 ^a 33.2 ^a 32.7 ^c 33.1 ^d	40.6 ^a 39.2 ^a 38.6 ^c 39.1 ^d	44.7 ^a 43.6 ^a 43.2 ^c 43.6 ^d	50.7 ^a 50.6 ^a 50.1 ^c 50.6 ^d
C ₄ H ₇ 1-4	19.7	24.1	28.3	32.0	37.7	42.1	49.2	20.1 ^a 20.3 ^c 20.7 ^d	25.0 ^a 25.0 ^c 25.3 ^d	29.2 ^a 29.4 ^c 29.5 ^d	32.8 ^a 33.2 ^c 33.1 ^d	38.6 ^a 39.0 ^c 38.7 ^d	42.9 ^a 43.4 ^c 43.0 ^d	49.7 ^a 50.2 ^c 49.8 ^d
C ₄ H ₅ -I	18.2	22.2	25.8	28.8	33.2	36.6	41.9	18.8 ^a 18.5 ^c 18.8 ^d	23.1 ^a 22.6 ^c 23.1 ^d	26.6 ^a 26.2 ^c 26.6 ^d	29.5 ^a 29.2 ^c 29.5 ^d	33.9 ^a 33.6 ^c 33.9 ^d	37.2 ^a 36.9 ^c 37.2 ^d	42.4 ^a 41.9 ^c 42.3 ^d
C ₄ H ₅ -N	18.1	22.7	26.7	29.9	34.0	37.0	41.6	18.7 ^a 17.8 ^c 18.7 ^d	23.8 ^a 21.7 ^c 23.7 ^d	27.7 ^a 25.2 ^c 27.7 ^d	30.7 ^a 28.2 ^c 30.7 ^d	34.9 ^a 32.6 ^c 34.8 ^d	37.7 ^a 35.9 ^c 37.7 ^d	42.2 ^a 41.0 ^c 42.1 ^d

Superscripts ^a, ^c and ^d are the same as in Table 2.

higher respectively. It can also isomerize to form C₄H₇1-OO4 radical after overcoming a relatively high energy barrier of 31 kcal/mol.

Similarly, eliminations of the other hydroperoxyl butenyl radical, C₄H₆1-4OOH3 shown in Fig. 1(a), can proceed through cyclization reactions to form (CYOC)C=C + $\dot{\text{O}}\text{H}$ or HOO(CYCC)C• radical, with lower barrier heights. It can also isomerize to produce C₄H₇1-OO3 radical or decompose to C₂H₃ + C=COOH, while barrier height for the latter is much higher.

3.2. Relative energies calculated using different methods

A comparison of the relative energies of important TSs and entrance wells to the reactants of C₄H₆ + HO_2 , calculated by using six different methods as described in Section 2, with the results from CBS-QB3 methodology reported by Miyoshi [13], is illustrated in Table 1.

Generally, the calculated relative energies of wells are in better agreement than the ones of TSs. Large discrepancy can be observed between the results of our ROCCSD(T)/CBS calculations and

those calculated by Miyoshi. Discrepancy for TS3, W3 and W4 is ~2 kcal/mol, while for TS5 and TS6 is up to ~5 kcal/mol. However, we obtained similar results as Miyoshi provided in the paper when using the same method of CBS-QB3, all within 0.2 kcal/mol except for one (0.8 kcal/mol). The different geometry optimization procedures inbuilt in the composite methods will eventually lead to different molecular structures employed in the single-point energy calculations, for example, G4 uses B3LYP/6-31G(2df,p) for geometry optimization and frequency computation whereas CBS-QB3 utilizes B3LYP/6-311G(2d,d,p) for these processes. While geometries remain the same (in this study, from BHandHLYP/6-311++G(d,p) method) throughout the single-point energy calculations when using ROCCSD(T)/CBS and CCSD(T)/MP2/CBS methods. Therefore, the differences in geometry can be a cause of the discrepancies in energy observed in Table 1 result from different methods employed.

Among the different methods displayed in Table 1, energies obtained at ROCCSD(T)/CBS and CCSD(T)/MP2/CBS levels of theories have a better agreement between each other. Relative energies obtained for the wells from these two methods are within

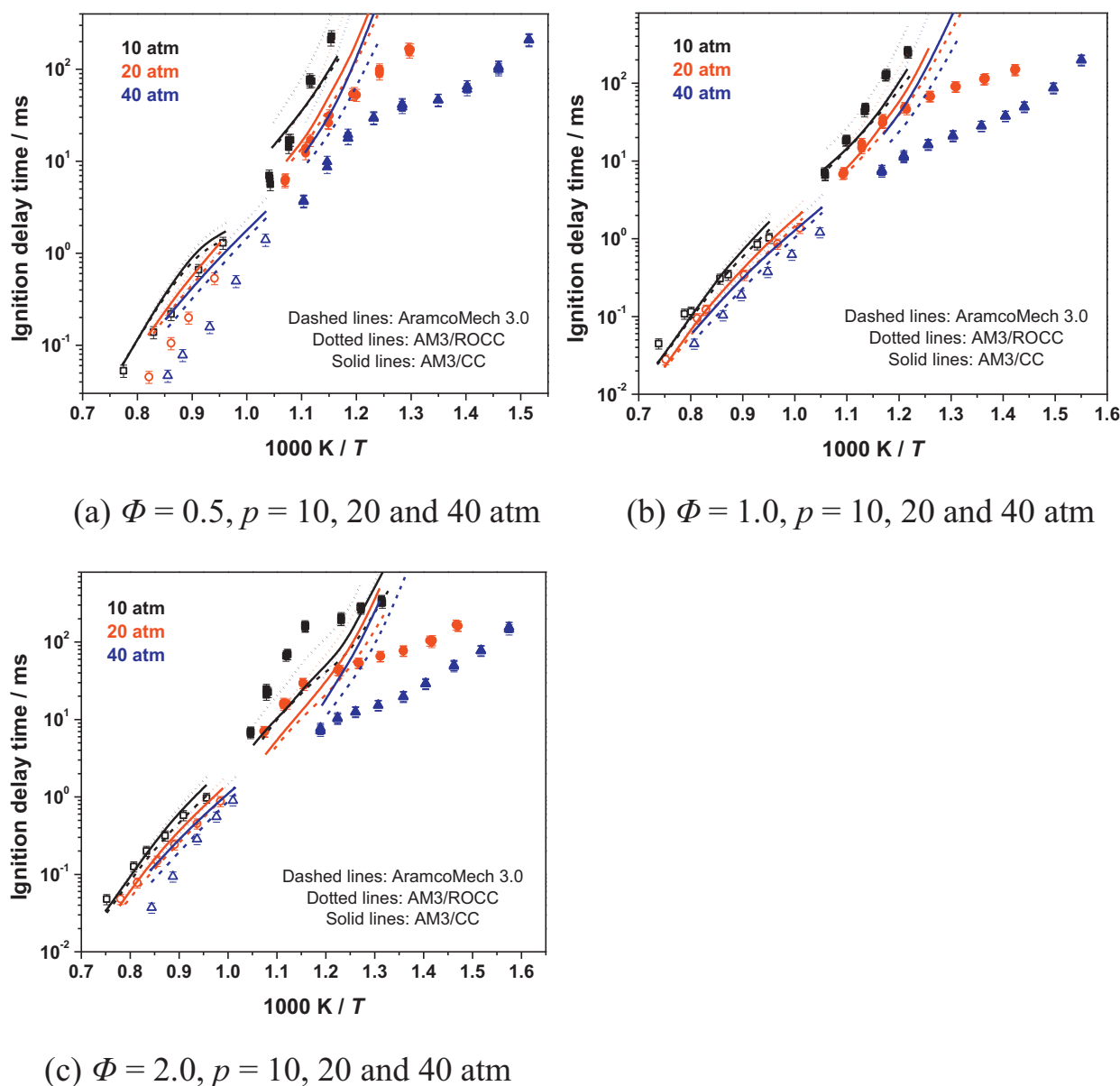


Fig. 4. IDTs for the C_4H_6 /air mixture at $p = 10, 20$ and 40 atm. Hollow symbols: NUIG ST measurements [9], solid symbols: NUIG RCM measurements [9]. Lines at higher temperatures result from constant volume simulations, while those at lower temperatures are simulation results accounting for facility effects.

0.6 kcal/mol, while the ones for TSs are within 2 kcal/mol. Such differences, however, may still have large impact on kinetic analyses. Therefore, the barrier heights of TSs 1, 2, 4 and 5 for the important entrance channels, obtained from ROCCSD(T)/CBS and CCSD(T)//MP2/CBS methodologies, were both employed in rate constant calculations for the reactions of 1,3-butadiene + HO_2 , while relative energies of the other stationary points were all calculated from method of ROCCSD(T)/CBS.

3.3. Rate constants and branching ratios

The pressure- and temperature-dependent rate constants calculated for the important reaction pathways of HO_2 addition to 1,3-butadiene, at temperatures of 500–2000 K and pressures of 0.01, 1 and 100 atm respectively, and the corresponding branching ratios for different reaction pathways in the same conditions are depicted in Fig. 2.

At 0.01 atm, the well-stabilization pathways are overwhelmed by the bimolecular elimination pathways in the entire tempera-

ture range from 500 to 2000 K. The formation of $(CYOC)C=C + \dot{O}H$ through terminal addition reaction pathways is the most prevalent over 700 K, while the relative yield of $C_4H_7-3 + O_2$ declines with increasing temperature and finally overwhelmed by the relative production of $(CYOC)C=C + \dot{O}H$ from central addition reaction pathways. At atmospheric pressure, the yields of resonantly stabilized $C_4H_6-3OOH4$ radical and C_4H_7-1OO3 radical become more important at temperatures below 800 K. At 100 atm, the stabilizations of C_4H_7-1OO3 and $C_4H_6-3OOH4$ dominate at temperatures below 800 K as expected, given that the wells of these two radicals are rather low-lying (-16.1 and -13.8 kcal/mol relative to the reactants, respectively). The well stabilization pathways are in competition with the bimolecular elimination pathways until 1100 K.

Energy barrier heights for the entrance channels influence their rate constants significantly, hence rate constants based on the barrier heights obtained at the CCSD(T)//MP2/CBS level of theory are also shown in Fig. 2 as dashed lines. Rate constants for the formation of $C_4H_6-3OOH4$ have increased by over an order of magnitude at 0.01 atm, and by around a factor of 5 at pressures over

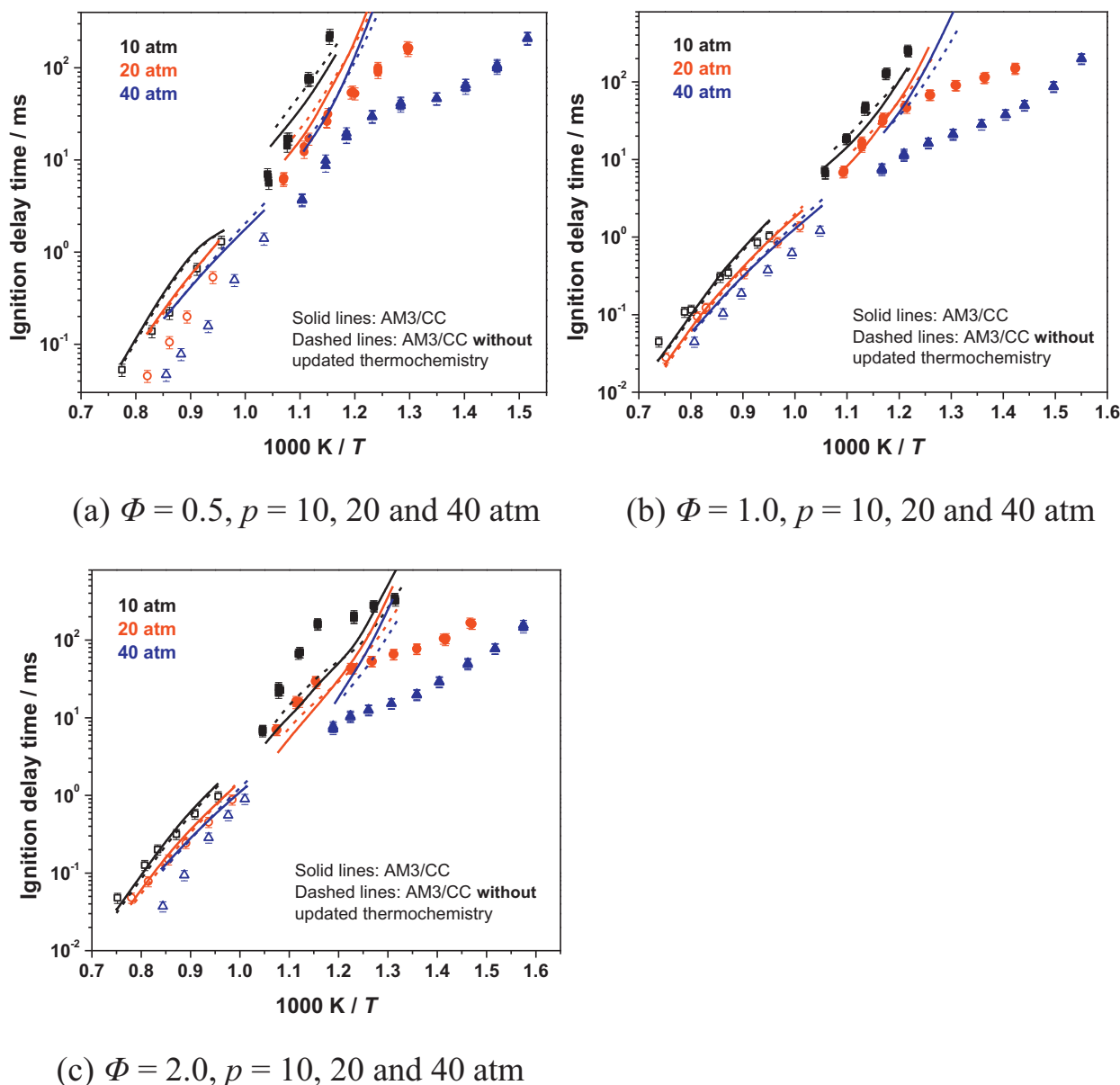


Fig. 5. Influence of thermochemistry data calculated in this study on IDTs for the C_4H_6 /air mixture at $p = 10, 20$ and 40 atm. Hollow symbols: NUIG ST measurements [9], solid symbols: NUIG RCM measurements [9]. Lines at higher temperatures result from constant volume simulations, while those at lower temperatures are simulation results accounting for facility effects.

1 atm. While the other reaction pathways generally accelerate by a factor of 1.2–3 at the same conditions. Accordingly, branching ratios have changed, with the formation of $(CYOC)C=C + \dot{O}H$ through terminal addition reaction pathways and that of $C_4H_6-3OOH_4$ being enhanced. Whereas the relative importance of the stabilization of $C_4H_7-1-OO_3$ and the formation of $C_4H_7-1-3 + O_2$ declines. Modified Arrhenius expressions for the rate constants of the important reaction pathways of 1,3-butadiene + $\dot{H}O_2$, based on the energy barriers from method of CCSD(T)//MP2/CBS, are provided in the Supplementary material.

The energetics of $\dot{H}O_2$ abstracting H-atom from different sites of 1,3-butadiene was determined by using method of ROCCSD(T)/CBS. Energy barrier for H-atom abstraction from the allylic-vinyl site (black circle in Fig. 3) by $\dot{H}O_2$ radical is 23 kcal/mol, while barrier heights for $\dot{H}O_2$ radical abstracting the two vinylic H atoms are identical (25.7 kcal/mol). High pressure limiting rate constants of H-atom abstraction reactions from 1,3-butadiene by $\dot{H}O_2$ were obtained through Transition State Theory (TST) calculations in MESS

packages, and compared with the total rate constants for $\dot{H}O_2$ addition reactions to 1,3-butadiene as shown in Fig. 3.

3.4. Thermochemistry

Temperature-dependent thermochemical properties at temperatures in the range of 298–3000 K for important C_4 species on the PESs of 1,3-butadiene + $\dot{H}O_2$ have been obtained through ab initio calculation as described in Section 2. The thermochemistry data for important C_4 species on the potential energy surfaces of 1,3-butadiene + $\dot{H}O_2$ calculated in this work are compared with the values from AramcoMech 3.0 [9], the Active Thermochemical Tables (ATcT) [40], the Third Millennium Ideal Gas and Condensed Phase Thermochemical Database for Combustion (Burcat's database) and the calculations by Goldsmith et al. [41], shown in Tables 2 and 3.

Thermochemistry data for most of the species are in good agreement with the ones in the literature. It can be seen that the

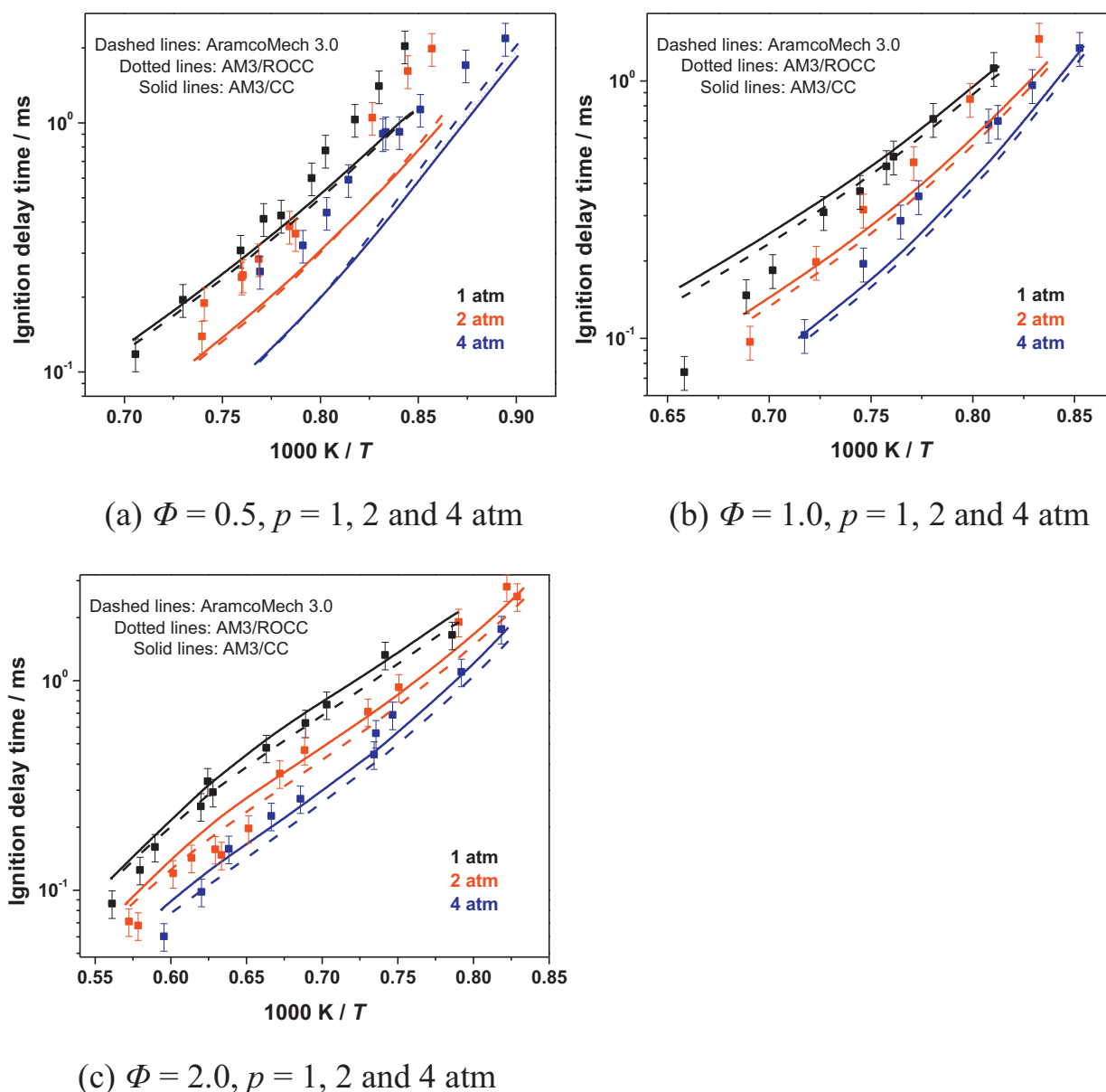


Fig. 6. IDTs for $C_4H_6/O_2/Ar$ mixture at $p = 1, 2$ and 4 atm. Symbols in (a): TAMU ST measurements [9], (b): KAUST ST measurements [9] and (c): XJTU ST measurements [9].

calculated standard-state enthalpy of formation ($\Delta H_{f,298}^\ominus$) for 1,3-butadiene is ~ 1 kcal/mol higher than the benchmark value from the ATcT database and other values from the literature. This is from the 0 K formation enthalpy of 31.1 kcal/mol calculated by a combination of G3/G4/CBS-QB3/CBS-APNO methodologies, which is 1.2 kcal/mol higher than the value in ATcT database [40]. The calculated standard-state enthalpies of formation for other species are in better agreement with those from the calculation by Goldsmith et al. [41] at QCISD(T)/CBS level of theory (within ~ 1 kcal/mol), than with the values from Burcat's database (within ~ 1.5 kcal/mol) and AramcoMech 3.0 (within ~ 1.5 kcal/mol). Our calculations show that the $S_{298}^\ominus$ for most species in Table 2 agree well with the literature data (within ~ 0.5 cal/mol/K), larger discrepancy of 2.9 cal/mol/K can be observed for $C_4H_7-OO_3$. Moreover, enthalpy of formation and heat capacities (C_p) for $C_4H_7-OO_3$ from our calculation also exhibit larger differences compared with the values in AramcoMech 3.0, with $\Delta H_{f,298}^\ominus$ differs by 1.6 kcal/mol and C_p differs by up to ~ 3.5 cal/mol/K at temperatures from 500 to 1000 K. Thermochemistry for $C_4H_7-OO_3$ in AramcoMech 3.0 is

calculated from group additivity which relies largely on the accuracy of the values assigned for functional groups, and may causes that differential. The thermochemistry data calculated in this study should be an improvement over those estimated based on group additivity. NASA polynomials for the thermochemistry data from our calculations are provided in the Supplementary material.

3.5. Model applications

Reactions of 1,3-butadiene + HO_2 are critical in determining the fuel reactivity of 1,3-butadiene at low- to intermediate- temperatures [9]. Rate constants and thermochemistry data for the system of 1,3-butadiene + HO_2 in AramcoMech 3.0 have been substituted by the newly computed ones in this study. Furthermore, an important bimolecular reaction pathway, $C_4H_6 + HO_2 \rightleftharpoons (CYOC)C=C + OH$, was added into AramcoMech 3.0, with contributions on the yields of products from both terminal and central addition reactions.

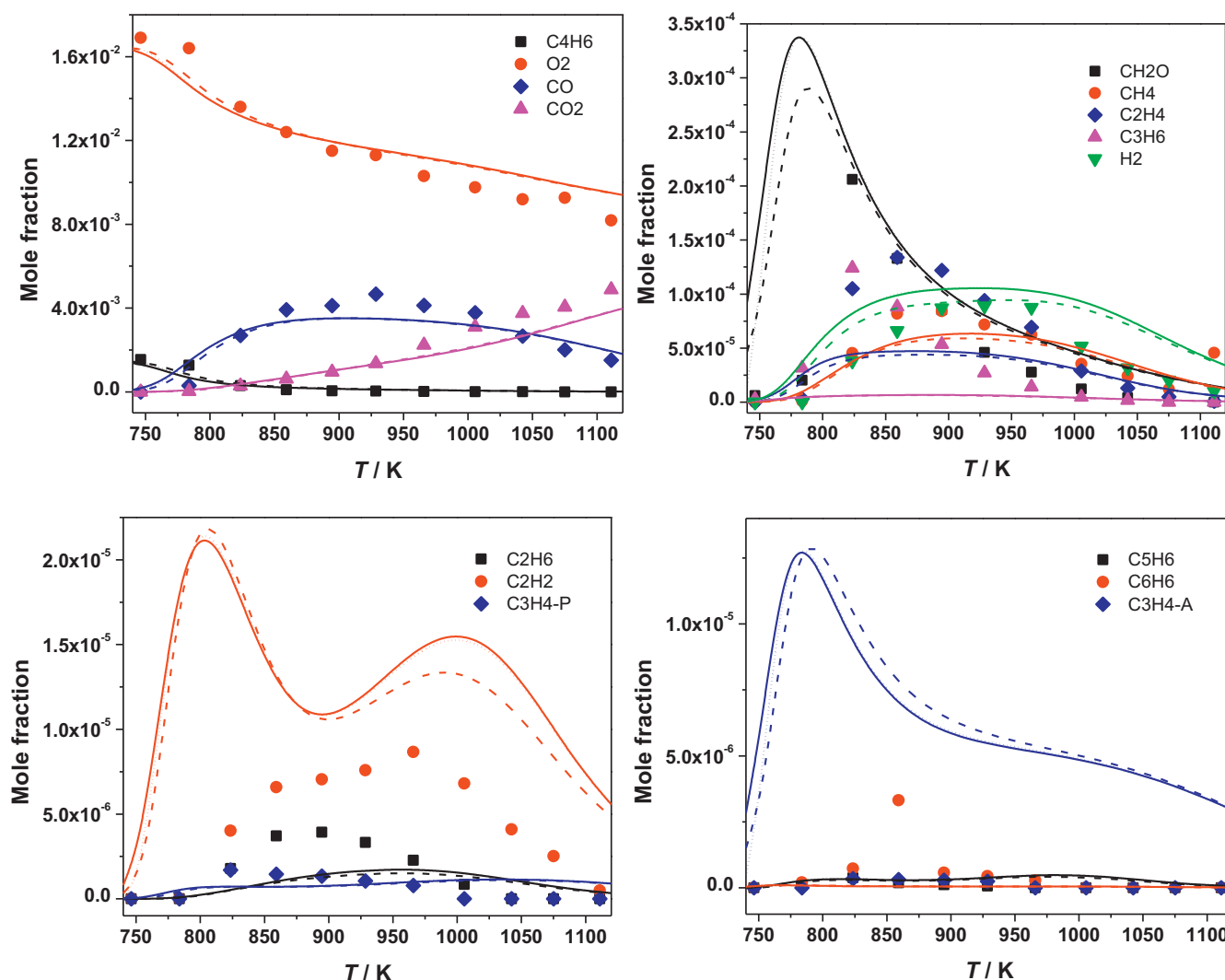


Fig. 7. 0.15% 1,3-butadiene, 1.65% O₂, 98.2% N₂, $\Phi = 0.5$, $p = 10$ atm, $\tau = 1$ s. Symbols: JSR measurements by Dagaut et al. [44], dashed lines: AramcoMech 3.0, dotted lines: AM3/ROCC and solid lines: AM3/CC.

Three mechanisms (AramcoMech 3.0, AM3/ROCC and AM3/CC) were used in our simulations for comparison. AM3/ROCC represents AramcoMech 3.0 with the rate constants for each individual reaction pathway related to 1,3-butadiene + HO₂ calculated by using the barrier heights from ROCCSD(T)/CBS methodology, while AM3/CC represents AramcoMech 3.0 adopting the rate constants calculated by using the barrier heights from CCSD(T)/MP2/CBS methodology. In AM3/ROCC and AM3/CC, the thermochemistry data for species shown in Tables 2 and 3 have been updated. The autoignition of 1,3-butadiene in shock tube was simulated under constant volume, homogeneous consumptions, while the volume profiles provided in [9] were used for the rapid compression machine (RCM) simulations to account for heat loss. Speciation of 1,3-butadiene oxidation in a jet-stirred reactor (JSR) measured by Dagaut and Cathonnet [44] and in a flow reactor measured by Laskin et al. [7] has also been simulated. All simulations were performed using Cantera [45].

3.5.1. Ignition delay time simulations

The simulated IDTs for 1,3-butadiene/air mixture at equivalence ratios of 0.5, 1.0 and 2.0, at 10, 20 and 40 atm, using the three different mechanisms, are illustrated in Fig. 4. Incorporations of the newly calculated rate constants based on ROCCSD(T)/CBS

energy barriers and thermochemistry data for the system of 1,3-butadiene + HO₂ into AramcoMech 3.0 (i.e. AM3/ROCC) increase the IDTs. While at intermediate temperatures AM3/CC gives better agreement with the experimental measurements for $\Phi = 0.5$ and 1.0 at 10 and 20 atm. It is clearly revealed in Fig. 4 that both of the two mechanisms, AM3/ROCC and AM3/CC, increase the ignition delay times predictions especially at elevated pressures, with fuel reactivity predicted by AM3/ROCC declines more dramatically compared with the AramcoMech 3.0 results.

In order to test the effects of the thermochemistry data from our calculation on model prediction, a comparison between the predicted IDTs from AM3/CC with and without updated thermochemistry data is shown in Fig. 5. It can be seen that without adopting the updated thermochemistry data into AM3/CC the predicted IDTs are increased by ~40% at intermediate to high temperatures in the range of 860–1050 K for all three equivalence ratios. While at temperatures above 1050 K, the IDTs predicted by AM3/CC without updated thermochemistry are slightly decreased by ~8%, and the same trends can be observed at temperatures below 840 K but more noticeable. The ST IDTs of 1,3-butadiene/O₂ mixture diluted in argon at $\Phi = 0.5$, 1.0 and 2.0 at 1, 2 and 4 atm simulated using AramcoMech 3.0, AM3/ROCC and AM3/CC showed in Fig. 6 also displayed the effects of updated thermochemistry. At

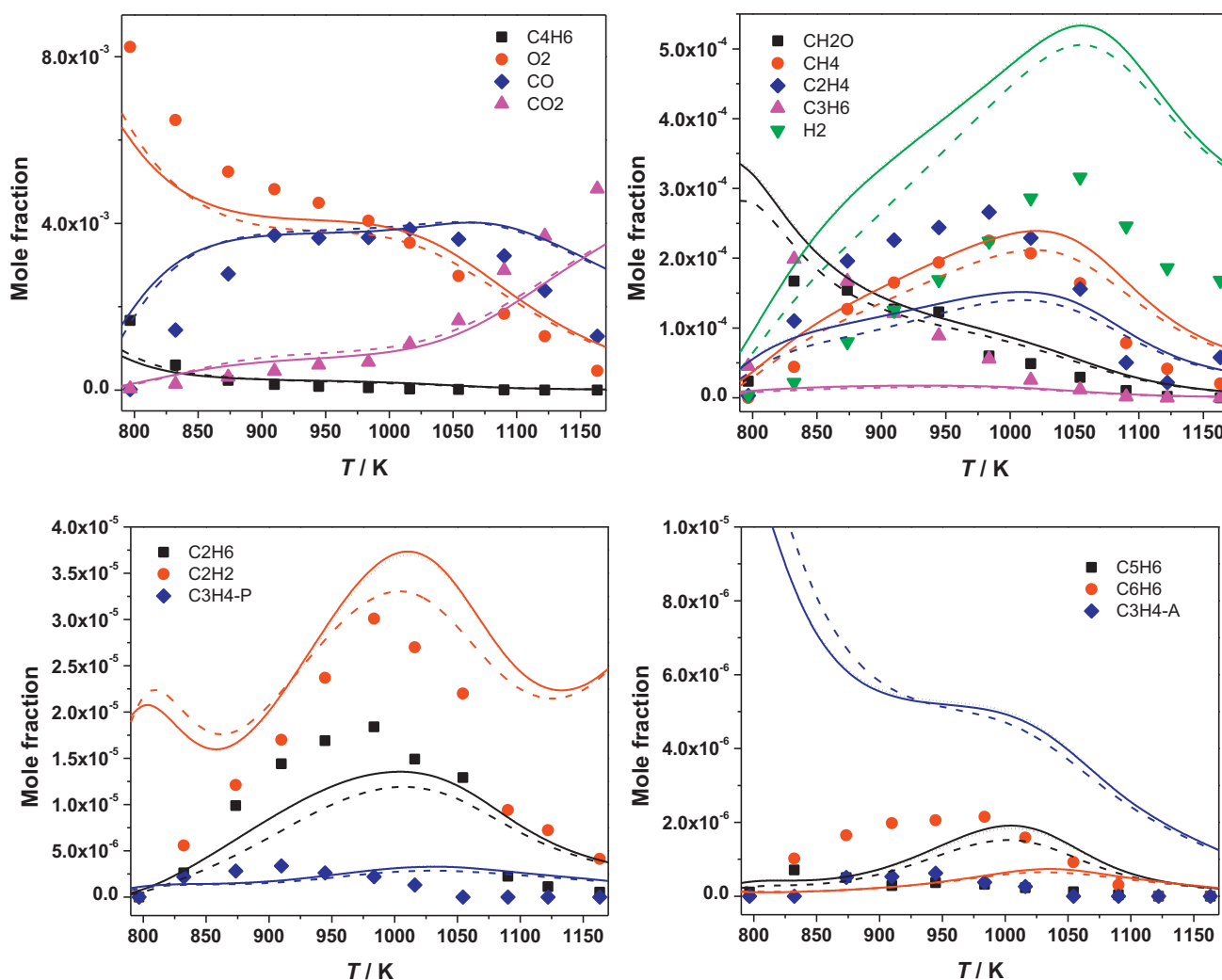


Fig. 8. 0.15% 1,3-butadiene, 0.825% O₂, 99.025% N₂, $\Phi = 1.0$, $p = 10$ atm, $\tau = 1$ s. Symbols: JSR measurements by Dagaut et al. [44], dashed lines: AramcoMech 3.0, dotted lines: AM3/ROCC and solid lines: AM3/CC.

temperatures above 1250 K under around atmospheric conditions, the results from AM3/ROCC and AM3/CC are almost superimposed where the kinetics of 1,3-butadiene + HO₂ plays a minor role, and the updated thermochemistry data in AM3/ROCC and AM3/CC lifts the predicted IDTs by ~10%.

This study only focuses on the reaction system of 1,3-butadiene + HO₂, other important reaction classes determining the fuel reactivity should be identified by sensitivity analyses on the new model, and will be further investigated along the same lines as this study.

3.5.2. Speciation simulations

The measured speciation data for 1,3-butadiene oxidation in JSR at equivalence ratios of 0.5 and 1.0 at 10 atm in the temperature range of 740–1160 K [44] were chosen as targets to simulate using AramcoMech 3.0, AM3/ROCC and AM3/CC, shown in Figs. 7 and 8. It can be seen that at temperatures above 800 K for both the conditions in Figs. 7 and 8 the difference between species profiles result from AM3/ROCC and AM3/CC is negligible, which indicates that the contribution from reaction kinetics of 1,3-butadiene and HO₂ radical to the destruction of fuel and O₂ and the formation of CO, CO₂ and other stable intermediates detected should be marginal. The updated thermochemistry data in AM3/ROCC and AM3/CC may lead to the difference in predicted species profiles

compared with AramcoMech 3.0. The mole fractions for CH₂O, CH₄, C₂H₄, C₂H₆ and H₂ predicted by AM3/ROCC and AM3/CC consistently increase at $\Phi = 0.5$ and 1.0 at 10 atm compared with AramcoMech 3.0 results. The consumptions of C₄H₆, O₂ are slightly increased at temperature below 850 K in AM3/ROCC and AM3/CC predictions.

Time profiles for different stable species during 1,3-butadiene oxidation in flow reactor at equivalence ratios of 0.55 and 1.63 at 1035 K and atmospheric pressure [7] have also been simulated using the three mechanisms, shown in Figs. 9 and 10. Again, the results predicted using AM3/ROCC and AM3/CC are almost the same indicating the minor influence of the kinetics of title reaction system on speciation. While changes in the time evolution profiles for other species from AM3/ROCC and AM3/CC predictions are not significant compared to AramcoMech 3.0 results, the profiles of C₄H₆ are slightly qualitatively improved as can be seen in Figs. 9 and 10.

3.5.3. Rate of production analyses

In order to examine the detailed chemistry changes with the three different mechanisms, rate of production analyses have been carried out for C₄H₆/air mixture using each mechanism at equivalence ratio of 0.5 at 870 K at 10, 20 and 40 atm respectively, and at equivalence ratio of 2.0 at 870 K and 20 atm. Fluxes were obtained at the time when 20% of fuel was consumed, shown in

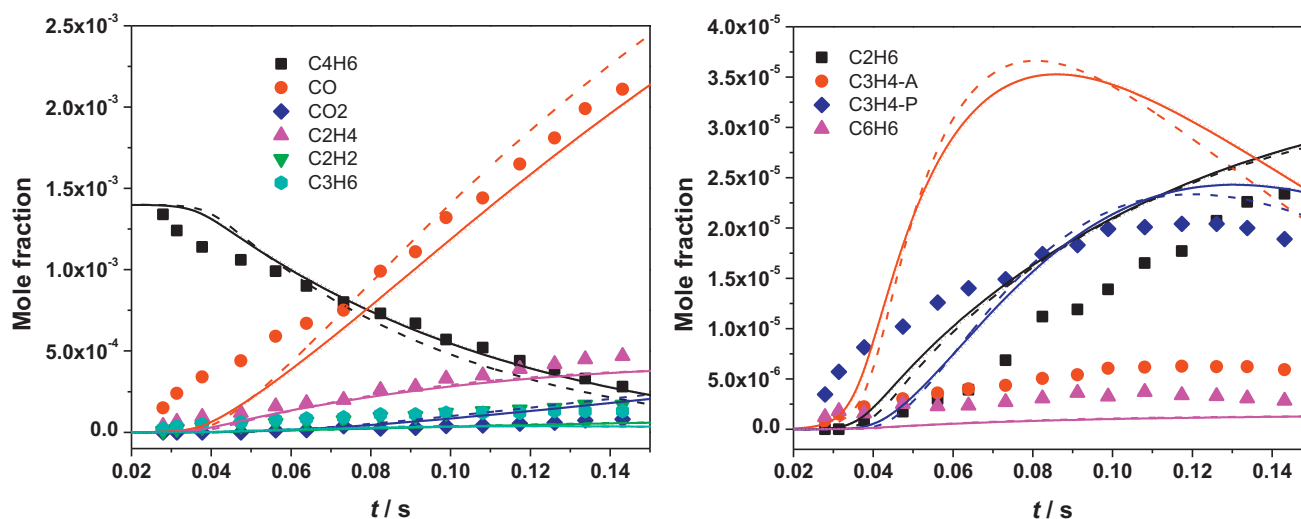


Fig. 9. 0.14% 1,3-butadiene, 1.4% O₂, 98.46% N₂, $\Phi = 0.55$, $p = 1$ atm, $T = 1035$ K. Symbols: flow reactor measurements by Laskin et al. [7], dashed lines: AramcoMech 3.0, dotted lines: AM3/ROCC and solid lines: AM3/CC.

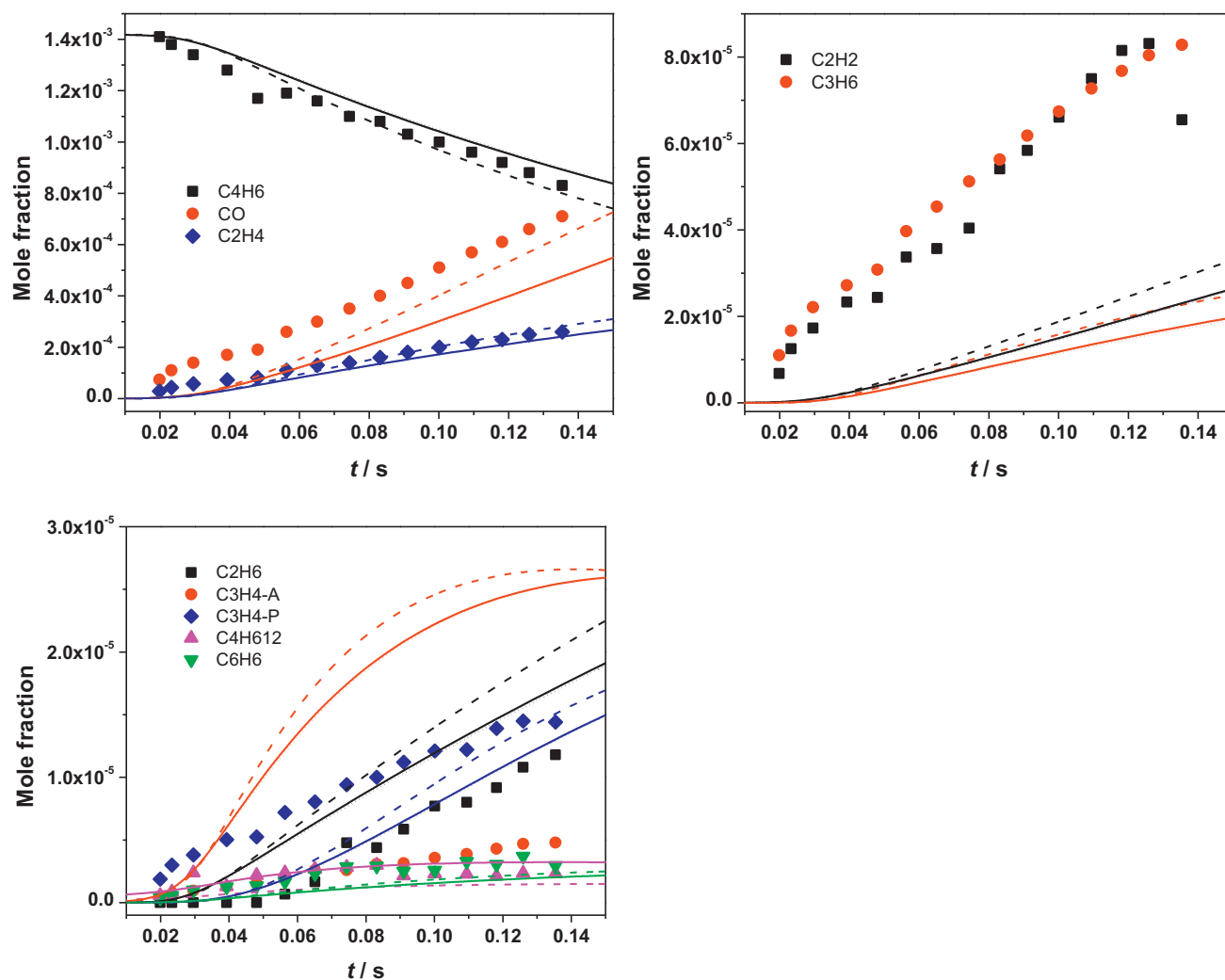


Fig. 10. 0.142% 1,3-butadiene, 0.48% O₂, 99.378% N₂, $\Phi = 1.63$, $p = 1$ atm, $T = 1035$ K. Symbols: flow reactor measurements by Laskin et al. [7], dashed lines: AramcoMech 3.0, dotted lines: AM3/ROCC and solid lines: AM3/CC. (Time shift: -0.02 s.).

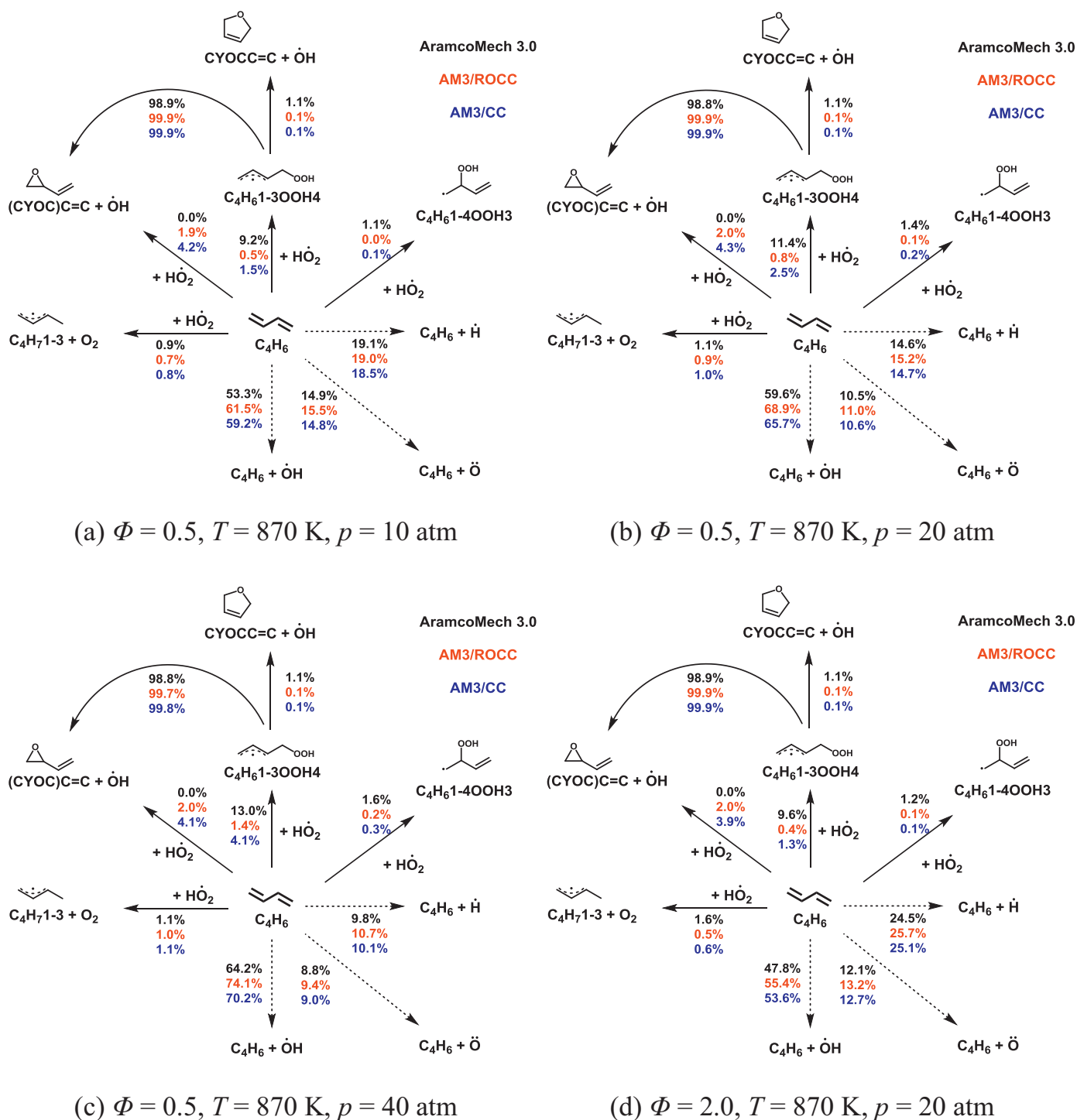


Fig. 11. Rate of production analyses for C_4H_6 /air mixture at 20% fuel consumption.

Fig. 11. Fluxes for fuel consumption in Fig. 11(a)–(c) correspond to the autoignition conditions of C_4H_6 /air mixture illustrated in Fig. 4(a), where differences among the predicted IDTs by AramcoMech 3.0, AM3/ROCC and AM3/CC increase with pressure. While the flux analysis shown in Fig. 11(d) corresponds to the autoignition condition illustrated in Fig. 4(c).

Given that the kinetics and thermochemistry data for 1,3-butadiene + HO_2 in AM3/ROCC and AM3/CC has been updated, major difference in chemistry among the three mechanisms displays in the branching fractions for $C_4H_6 + HO_2$. Since bimolecular reaction pathway of $C_4H_6 + HO_2 \leftrightarrow (CYOC)C=C + \dot{O}H$

was not included in AramcoMech 3.0 and the production of $(CYOC)C=C + \dot{O}H$ totally from the cyclization reaction of $C_4H_6-3OOH_4$ radical. While for AM3/ROCC and AM3/CC, the formation of $(CYOC)C=C + \dot{O}H$ dominates in reaction system of $C_4H_6 + HO_2$ at 10 atm and starts to compete with the formation of $C_4H_6-3OOH_4$ radical as pressure goes up, show in Fig. 11(a)–(c). Although the total flux of $C_4H_6 + HO_2$ for fuel consumption is rather small compared with the contributions from $C_4H_6 + \dot{O}H$, $C_4H_6 + \dot{H}$ and $C_4H_6 + \dot{O}$ reactions, the major chain propagation reaction pathways in $C_4H_6 + HO_2$ reaction system provides highly reactive $\dot{O}H$ radical and thus significantly accelerates autoignition. The total

flux of $C_4H_6 + \dot{O}H$ has been indirectly affected by the modifications in $C_4H_6 + \dot{H}O_2$ chemistry in AM3/ROCC and AM3/CC and varies by up to 9.9% compared with AramcoMech3.0 at the four conditions in Fig. 11(a)–(d), whereas the total fluxes of $C_4H_6 + \dot{H}$ and $C_4H_6 + \dot{O}$ stay similar for the three mechanisms, within 1.2%. Therefore, the rate of production analyses here also highlights the importance of $C_4H_6 + \dot{O}H$ reaction system in determining 1,3-butadiene fuel reactivity at intermediate temperatures, and since it will couple with the title reaction system and account for more than 50% of fuel consumption. Theoretical investigation on the reaction system of $C_4H_6 + \dot{O}H$ and its application to model prediction has been carried out in our recent work [46].

4. Conclusions

Chemical kinetics of 1,3-butadiene + $\dot{H}O_2$ has been studied in detail in this work, and its influence on model prediction has also been investigated. Zero-point energy corrected PESs for central and terminal addition reaction pathways have been obtained at ROCCSD(T)/CBS//BHandHLYP/6–311++G(d,p) level of theory. Comparison of ZPE corrected relative energies from six different computational methods for the important TSs and intermediates, with calculation results in the literature, indicates that uncertainty introduced from the method employed in SPE calculations is non-negligible. Energy barrier heights for the important entrance pathways determined from the ROCCSD(T)/CBS and CCSD(T)/MP2/CBS levels of theory were both used in kinetic analyses. By solving the RRKM/ME, pressure- and temperature-dependent rate constants at temperature range of 298.15–2000 K at pressures from 0.01 to 100 atm, for the reactions of $C_4H_6 + \dot{H}O_2$, have been obtained. Temperature-dependent thermochemical properties for important species of the title reactions have also been calculated.

As can be seen from the branching ratios of different reaction pathways, the formation of vinyl oxirane, (CYOC)=C, and $\dot{O}H$ radical dominates the entire temperature range of 500–2000 K at 1 atm and below; stabilizations of the initial adducts, 1-hydroperoxy-3-buten-2-yl radical ($C_4H_6-3OOH_4$) and 3-butenyl-2-peroxy radical ($C_4H_7-1OO_3$), start to be prevalent at elevated pressures and compete with the bimolecular elimination pathways.

The new kinetics data based on ROCCSD(T)/CBS and CCSD(T)/MP2/CBS energy barrier heights and the updated thermochemistry data, for reaction pathways of 1,3-butadiene + $\dot{H}O_2$, have been incorporated into AramcoMech 3.0. The ignition delay times of C_4H_6 /air and C_4H_6/O_2 /Ar mixtures, and the speciation of 1,3-butadiene oxidation in JSR and flow reactor have been simulated using three different mechanisms, AramcoMech 3.0, AM3/ROCC and AM3/CC. The updated kinetics data in this work tends to increase the predicted IDTs of C_4H_6 /air mixture at intermediate temperatures, with the predictions by AM3/ROCC being the longest. Effects of updated thermochemistry for 1,3-butadiene + $\dot{H}O_2$ on fuel reactivity and the formation of stable species have been tested. The updated thermochemistry data in this work slightly increases fuel reactivity and fuel/oxidizer consumption at intermediate temperatures. Rate of production analyses have been carried out for the three mechanisms. Although the total flux of $C_4H_6 + \dot{H}O_2$ for fuel consumption is rather small compared with the contributions from $C_4H_6 + \dot{O}H$, $C_4H_6 + \dot{H}$ and $C_4H_6 + \dot{O}$, the major chain propagation reaction pathways in $C_4H_6 + \dot{H}O_2$ reaction system provides highly reactive $\dot{O}H$ radical and thus significantly accelerates autoignition. The importance of $C_4H_6 + \dot{O}H$ reaction system in determining 1,3-butadiene fuel reactivity is also highlighted. Even though deviation between the simulations and experimental measurements still exists, the mechanism for 1,3-butadiene oxidation should be improved after adopting the rate constants and thermochemistry data of 1,3-butadiene + $\dot{H}O_2$ calculated in this study.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.07.033.

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